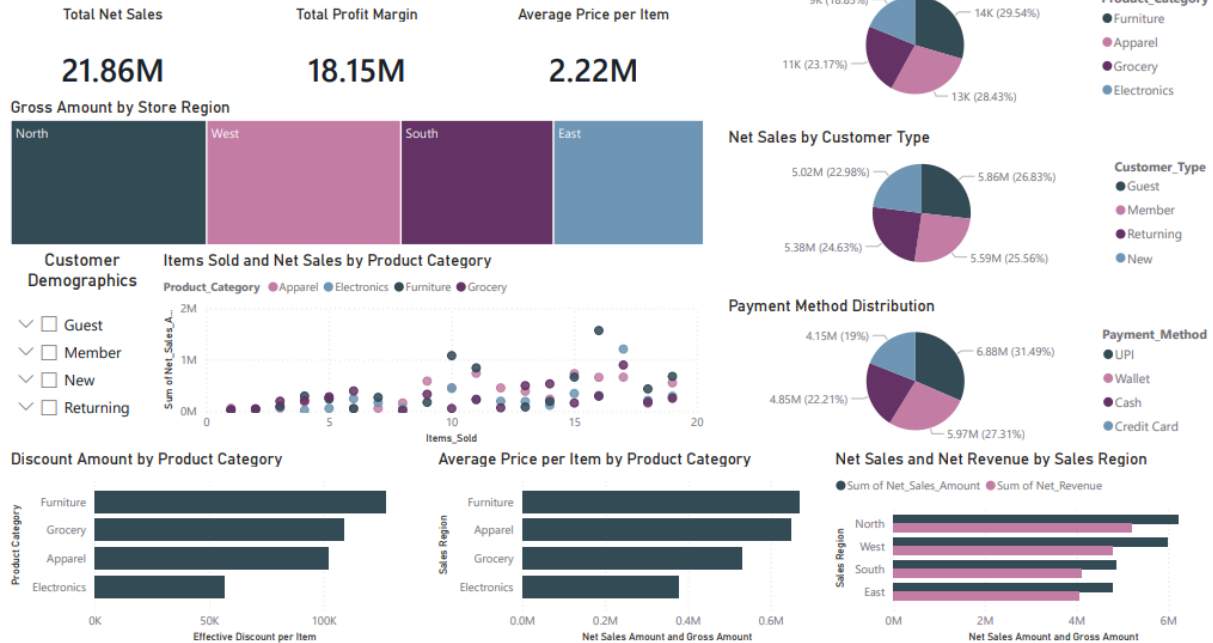


Retail Store Sales Dataset Report

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Retail Store Sales Dashboard



Introduction

The dataset used in this project contains data from retail store sales. It contains a number of categories that are crucial for determining sales performance such as price, product category, profit margin, customer demographics, store locations, and payment methods used by customers. To ensure the reliability of the gathered data, data cleaning was done to screen for missing values and inconsistent formatting. The main concern with uncleaned dataset is that it could provide misleading information and cause errors on the tools being used to analyze the dataset such as on Power BI. Creating appropriate visualizations would also prove to be difficult.

Data Issues Identified

When cleaning this specific dataset, it is found that there is a total of 1,032 missing values in the dataset. The attribute that had the most amount of missing values is the Effective_Discount_Per_Item at around 240. The use of the transaction ID for each column helped eliminate duplicate values on the dataset. There are also no data type issues that have been during the process of data cleaning.

Cleaning Decisions and Justification

In this particular dataset, there have been no data removal and a reasonable amount of imputation done to the missing attributes. Imputation would be most appropriate if the columns only lack a minimal amount of attributes that can easily be inferred such as Payment_Method or Product_Category, or calculated using existing values such as Discount_Percent or Average_Price_Per_Item. However, if there are multiple attributes missing from the row, it is

more appropriate to remove the data entirely to avoid introducing erroneous data from poor inference of data or causing errors on the tools used. Median is used to show the mid point of the statistical dataset. Median is best used if the values on the dataset could skew the mean/average of the dataset higher or lower than the median of the data. Standardizing is necessary in data analysis and visualization as it helps streamline the formatting of the attributes in the dataset as text capitalization and variable types could errors during the visualization process.

New Variables Created

Derived variables are attributes or variables in the dataset that have been created using two or more existing attributes in the dataset. They are useful to provide new information that could be more insightful than the existing variables already provide. Derived variables provide visualization readiness as there are values or variables in the dataset that on its own cannot be visualized such as the Store_Region attribute in this dataset; however, deriving a new variable that uses the count of each value in the column could allow it to be visualized into a pie chart or a tree map diagram. An example of a derived variable from our output is the Net Revenue which is the output of the Net Sales Amount subtracted towards the Discount Amount.

Aggregation & Validation

The following are the aggregations used in the dataset visualization:

- Net Sales by Product Category
- Net Sales by Customer Type
- Net Sales and Profit Margin by Sales Region
- Average Price per Item by Product Category
- Discount amount by Product Category
- Item Sold and Net Sales by Product Category
- Gross Amount by Store Region
- Total Net Sales
- Total Profit Margin
- Average price per item

Reflection

The data cleaning executed in this data has been successful and the steps in conducting the data cleaning procedure has been followed from finding missing values to standardizing the formatting of the dataset. This project has also managed to create and utilize derived variables from the dataset to create additional visualizations in the dashboard.